

WASP-52b

R-0333

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_221101 · created 2026-07-18T05:22:22+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459884.671 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.012
Transit depth (Rp/Rs)^2	3.14 +/- 0.44 [%]
Orbital Inclination (inc)	85.17 +/- 0.83
Airmass coefficient 1 (a1)	1.001 +/- 0.01
Airmass coefficient 2 (a2)	-0.0 +/- 0.055
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	#3 - [517, 235]
Optimal Method	PSF photometry
Transit Duration (day)	0.0755 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.177 ± 0.012 vs 0.1646 (deviation 1.03σ)
Duration fit vs published	0.0755 d vs 0.0758 d (deviation 0.06σ)
Mid-time O–C	1.6 min
Depth SNR	14.7
Residual scatter	1.02 %

Light curve

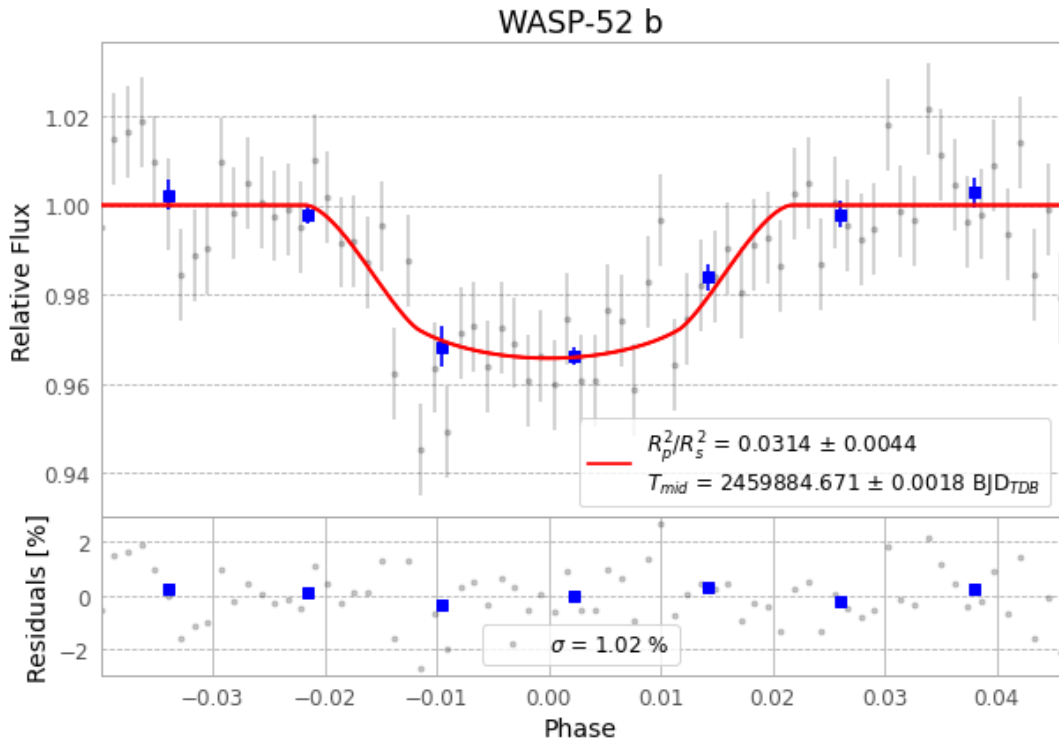


Figure 1 — FinalLightCurve_WASP-52 b_2022-10-31.png (SHA-256 e858bda83f31aab6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 153], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7d0292d51993fbbb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 061fc91

Data provenance

73 FITS frames (48 MB), WASP-52221101021815.FITS → WASP-52221101055415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-221101.log (a90bff556a98...), FinalLightCurve_WASP-52 b_2022-10-31.png (e858bda83f31...), FinalLightCurve_WASP-52 b_2022-10-31.csv (893d003dbaa2...)

Compute resources

Wall time 110.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 05:22Z.